OpenFOAM Final Project Proposal

Due date on Canvas

COE 347

There is absolutely no tolerance for academic misconduct. Late assignments will not be accepted, unless under exceptional circumstances at the instructor's full discretion.

Submit your final project electronically as a PDF via Canvas by 11:59pm on the due date.

Instructions

The class features a team-based final project requiring the students to design, plan, and execute a simulation using the `OpenFOAM` software.

The project must feature a canonical flow chosen between (a) flow above and inside a cavity; (b) laminar flow around an object; (c) supersonic flow around an object. More details are below.

A successful final project report will feature

- A clear and well posed engineering or physical question that you wish to answer using CFD.
- An informative and succinct summary of the quantitative behavior of the system via adequate graphs and plots of key variables.
- Whenever possible, comprehensive comparisons against data from previous studies.
- A careful and well-detailed analysis and appraisal of the errors and inaccuracies related to time steps and mesh spacings of finite size as to convince yourselves that your numerical solutions are trustworthy.

A final report based on the work performed within the project is to be turned in on the due date. Please keep the project report between 5 to 10 pages in length.

Projects and topics

In an effort to keep the workload manageable, you are encouraged to propose a project that builds on top and extends one of the three OF assignments.

For example, your group could consider the flow over cavities of irregular shapes, or the flow around a square or triangular cylinder, or flows around various objects in a supersonic wind-tunnel (including viscous forces at the wall).

When selecting a project, make sure to think carefully about the questions that you wish to answer or the flow features and key quantities that you wish to investigate.

Grading rubric

The following grading rubric will be used for the final project.

- Clear statement about objectives (20%)
- Technical complexity of project (20%)
- Overall quality and clarity of presentation of the material (20%)
- Comparisons against existing results and/or theory (20%)
- Thoroughness of the analysis, especially with regard to convergence (spatial and temporal as applica-ble) and evidence presented in support of the accuracy of the solutions (20%)